Light as an Electromagnetic Wave

The following subjects will be covered

- The EM Wave Equation and Solutions.
- The Refractive Index Relation.
- Light Wave Power Density.

Time-harmonic (Sinusoidal) Light Wave

Electric field Part of the wave

Amplitude

Phase

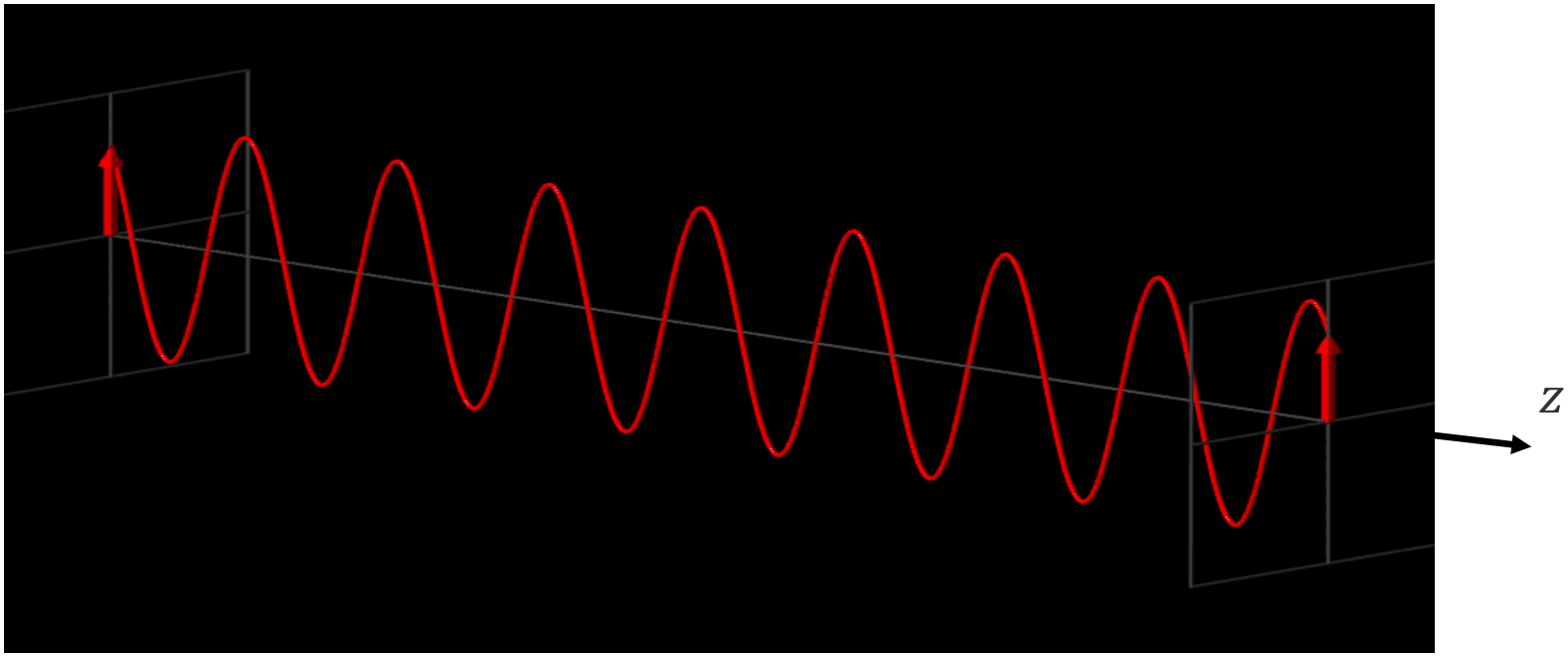
$$\vec{E}(z,t) = E_o \sin(\omega t \pm kz)\ \vec{a}_x$$

Field orientation/polarization

$\omega = 2\pi f$ Frequency

$k = \frac{\omega}{v}$ Wave number

Direction of wave propagation



The Uniform Plane-wave Solution

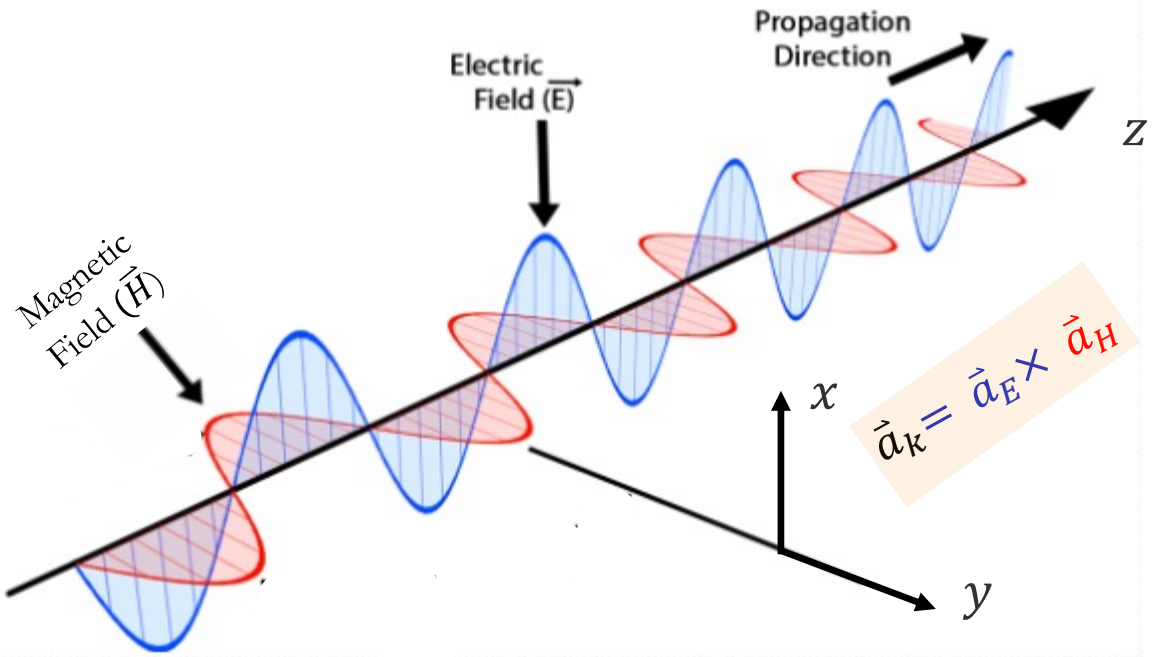
$$\vec{E}(z,t) = E_o \sin(\omega t - kz)\ \vec{a}_x$$

What about $\vec{H}$?

The EM fields are related using Maxwell's equations.

$$\vec{E}(z) = E_o\, e^{-jkz}\ \vec{a}_x$$

$$\vec{\nabla} \times \vec{E} = -\mu \frac{\partial \vec{H}}{\partial t}$$



For the 1D TH case, we have

$$\vec{H} = \frac{1}{-j\omega\mu}\ \vec{\nabla} \times \vec{E}\ = \frac{1}{-j\omega\mu}\ (-jk) E_o\, e^{-jkz}\ \vec{a}_y\ = \frac{k}{\omega\mu}\ E_o\, e^{-jkz}\ \vec{a}_y \qquad k^2 = \omega^2 \mu\varepsilon$$

$$\vec{H} = \sqrt{\frac{\varepsilon}{\mu}}\, E_o\, e^{-jkz}\, \vec{a}_y$$

$\sqrt{\frac{\mu}{\varepsilon}}$ has units of impedance (Ω)

$$\eta = \frac{E_o}{H_o} = \sqrt{\frac{\mu}{\varepsilon}}$$ Plane wave impedance

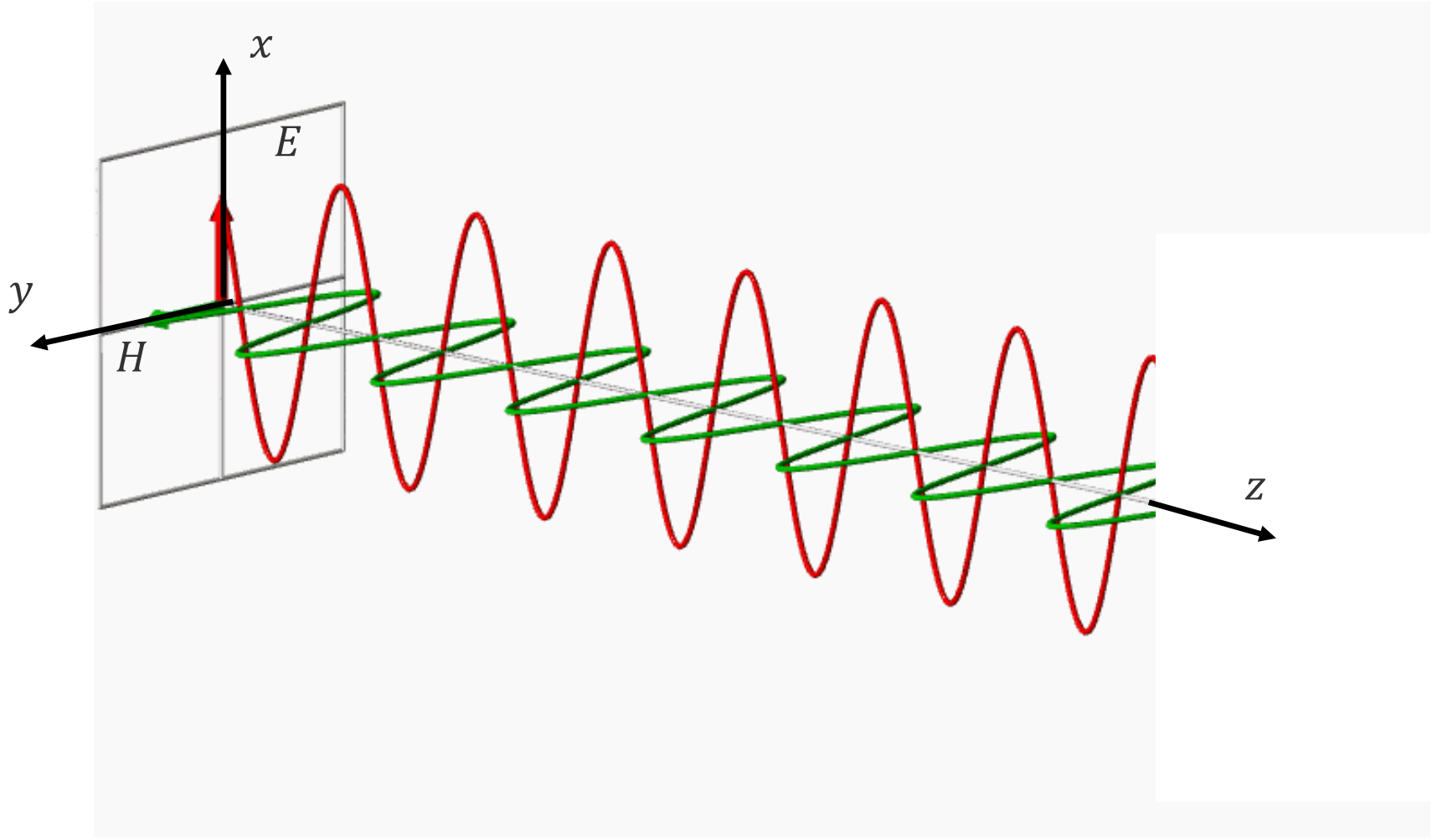
$$H_o = \frac{E_o}{\eta}$$

$$\vec{H} = H_o\, e^{-jkz}\ \vec{a}_y$$

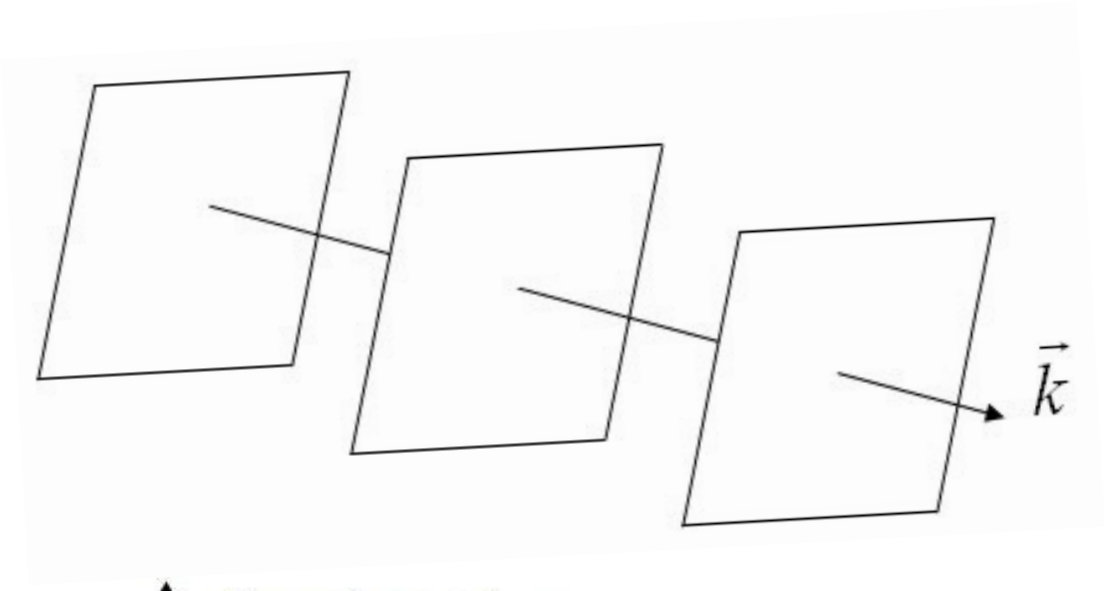
For free-space $\eta_o = \sqrt{\frac{\mu_o}{\varepsilon_o}} = 377\ \Omega$

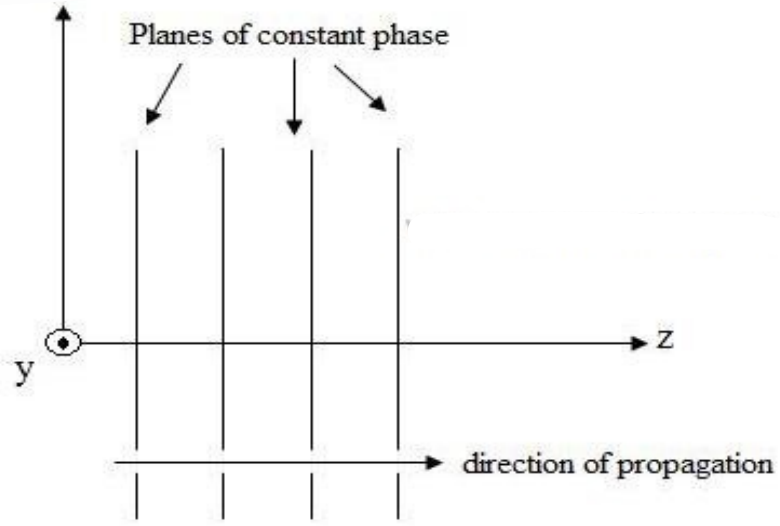
Other media $\eta = \frac{\eta_o}{n}$

The Uniform Plane-wave Solution



The Uniform Plane-wave Solution





Example

Consider an EM wave travelling in free space (n=1). The wave is polarized in the y-direction and is given by

$$\vec{E}(x,t) = 5\cos(2.7\times10^{15}t + kx)\ \vec{a}_y \quad \text{(V/m)}$$

The direction of propagation, $-ve\ x$ direction.

Frequency $\omega = 2.7\times10^{15}$ rad/s.

Phase constant (wavenumber) $k = \frac{\omega}{v} = \frac{2.7\times10^{15}}{3\times10^{8}} = 0.9\times10^{7}$ rad/m.

How many meters to complete a full 2π phase?

Wavelength $\lambda = \frac{v}{f} = 680$ nm (red light).

Write the expression for the magnetic field. $H_o = \frac{E_o}{\eta_o} = \frac{5}{377}$

$\vec{a}_k = \vec{a}_E \times \vec{a}_H$

$-\vec{a}_x = \vec{a}_y \times \vec{a}_H$

$$\vec{H}(x,t) = \frac{5}{377}\cos(2.7\times10^{15}t + kx)\ (-\vec{a}_z) \quad \text{(A/m)}$$

TH Plane-wave Solution in Lossy Media

Complex Refractive index

$$n = \sqrt{\mu_r \varepsilon_r} = n' - jn''$$

Complex propagation constant

$$k = \frac{n\omega}{c}$$

$$k = \frac{\omega}{c} n'' + j \frac{\omega}{c} n'$$

$$k = \alpha + jk$$

Field attenuation

Phase constant

$$\alpha = \frac{\omega}{c} n''$$

$$k = \frac{\omega}{c} n'$$

Refractive index

$$(n')^2 = \frac{\varepsilon_r'}{2} + \frac{1}{2}\sqrt{(\varepsilon_r')^2 + (\varepsilon_r'')^2}$$

Extinction coefficient

$$n'' = -\frac{\varepsilon_r''}{2n'}$$

Other relations

$$\varepsilon_r' = (n')^2 + (n'')^2$$

$$\varepsilon_r'' = -2n'\, n''$$

Wave speed

$$v = \frac{c}{n'}$$

Wave impedance

$$\eta = \frac{\eta_o}{n} = \frac{\eta_o}{n' - jn''}$$

The Plane-wave Solution in Lossy Media

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\,\vec{a}_x + E_y(x,y,z,t)\,\vec{a}_y + E_z(x,y,z,t)\,\vec{a}_z$$

Consider a one-dimensional case (z) with one component only.

$$\frac{\partial^2 E_x}{\partial z^2} - k^2 E_x = 0 \qquad k \text{ is complex}$$

$$E_x(z) = A\,e^{-kz} + B\,e^{kz}$$

Wave travelling along the $+ve\ z$ direction

Wave travelling along the $-ve\ z$ direction

Complex wave impedance

$$\eta = \frac{E_o}{H_o} = |\eta|\,\angle\theta_\eta$$

$$\vec{a}_k = \vec{a}_E \times \vec{a}_H$$

Take the $+z$ wave

$$E_x(z) = E_o\,e^{-(\alpha+jk)z}$$

$$E_x(z) = E_o\,e^{-\alpha z}\,e^{-jkz}$$

Attenuation

Phase

Take the sine wave solution

$$E_x(z,t) = E_o \sin(\omega t - kz)\,e^{-\alpha z}$$

$$\vec{E}(z,t) = E_o \sin(\omega t - kz)\,e^{-\alpha z}\,\vec{a}_x$$

$$\vec{H}(z,t) = H_o \sin(\omega t - kz)\,e^{-\alpha z}\,\vec{a}_y$$

Example

Ultraviolet sun light propagation in seawater. At UV wavelength (400 nm), the refractive index is $n = 1.34451 - j2.11\times10^{-10}$. If the magnetic field of this wave at some location inside seawater is given below, find the propagation parameters and the electric field expression.

$$\vec{H}(z,t) = 10\, e^{-\alpha z} \cos(\omega t - kz)\ \vec{a}_y \quad \text{(mA/m)}$$

The direction of propagation, $+ve\ z$ direction $\vec{a}_k = \vec{a}_z$.

$n' = 1.34451$

$$k_o = \frac{\omega}{c} = \frac{2\pi}{\lambda_o} = \frac{2\pi}{400\ nm} = 15.7\times10^6 \quad \text{rad/m}$$

$$k = k_o n' = (15.7\times10^6)(1.34451) = 21.12\times10^6 \quad \text{rad/m}$$

$n'' = 2.11\times10^{-10}$

$$\alpha = k_o n'' = (15.7\times10^6)(2.11\times10^{-10}) = 33.1\times10^{-4} \quad \text{1/m}$$

$$\eta = \frac{\eta_o}{n} \approx \frac{377}{1.34451} = 280.4 \quad \text{Ohms}$$

$$\vec{E}(z,t) = 2.804\ e^{-\alpha z} \cos(\omega t - kz)\ \vec{a}_x \quad \text{V/m}$$

Power Density in a Plane Wave

Consider the following Plane wave represented by E and H.

$$\vec{E} = E_o \cos(\omega t - kz)\, e^{-\alpha z}\ \vec{a}_x$$

$$\vec{H} = H_o \cos(\omega t - kz - \theta_\eta)\, e^{-\alpha z}\ \vec{a}_y$$

$$\vec{S} = \vec{E} \times \vec{H}$$

Poynting vector (Wave power density W/m^2)

$$\vec{S} = E_o H_o \cos(\omega t - kz)\cos(\omega t - kz - \theta_\eta)\ e^{-2\alpha z}\ \vec{a}_z$$

$$\vec{S} = \frac{{E_o}^2}{|\eta|}\cos(\omega t - kz)\cos(\omega t - kz - \theta_\eta)\ e^{-2\alpha z}\ \vec{a}_z$$

The time-average power density is given by

$$\vec{S}_{ave} = \frac{1}{T}\int_0^T \vec{S}\ dt\ \vec{a}_z$$

(W/m^2)

For a plane wave,

$$\vec{S}_{ave} = \frac{1}{2}\, Re\left[\vec{E} \times \vec{H}^*\right] \Rightarrow \vec{S}_{ave} = \frac{{E_o}^2}{2|\eta|}\cos\theta_\eta\, e^{-2\alpha z}\ \vec{a}_z$$

Total average power within some area

$$P_{ave} = \int_A \vec{S} \cdot \overrightarrow{dA} \quad \text{(W)}$$

Example

Consider an EM wave given by $\vec{E} = 4\ \sin(8.8\pi\times10^{14}t - kx)\ \vec{a}_z$ (V/m) travelling in a non-magnetic medium of n =3.8. Find k, η and $\vec{S}_{ave}$.

Lossless medium $\alpha = 0$

Frequency $\omega = 8.8\pi\times10^{14}$ rad/s

Refractive index $n = 3.8$ (real)

$$k = \frac{\omega}{c}n = \frac{8.8\pi\times10^{14}}{3\times10^{8}}(3.8) = 3.5\times10^{7} \text{ rad/m}$$

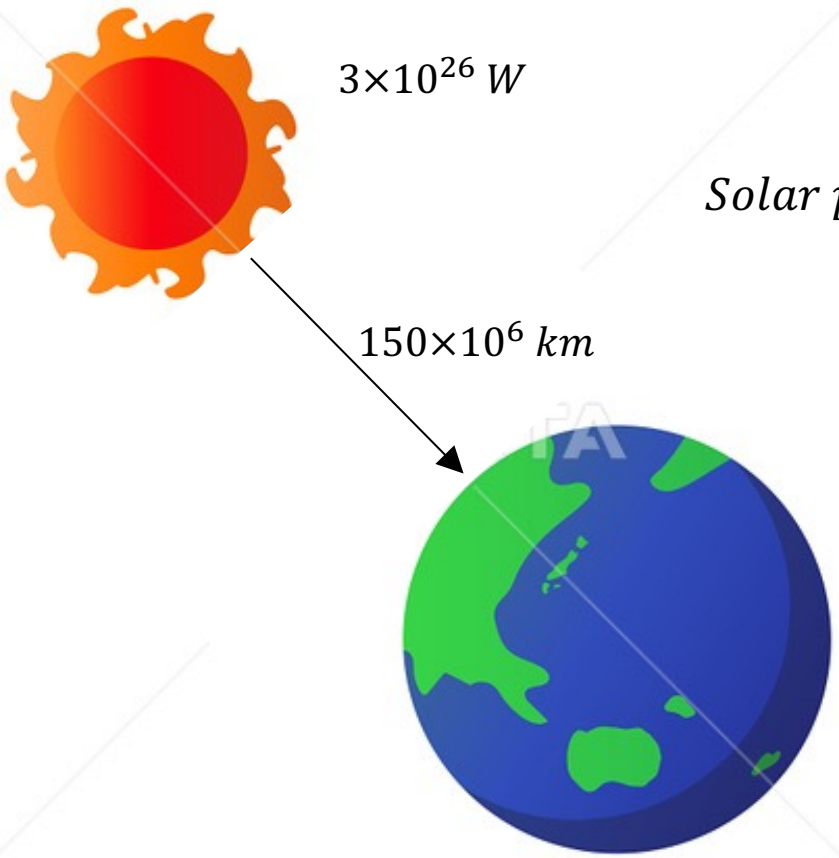
$$v = \frac{c}{n} = \frac{3\times10^{8}}{3.8} = 0.79\times10^{8} \text{ m/s}$$

$$\eta = \frac{\eta_o}{n} = \frac{377}{3.8} = 98.6\ \Omega$$

$$\vec{S}_{ave} = \frac{{E_o}^2}{2|\eta|}\cos\theta_\eta\ e^{-2\alpha z}\ \vec{a}_x$$

$$\vec{S}_{ave} = \frac{4^2}{2(98.6)}\ \vec{a}_x = 81.1\ \vec{a}_x \text{ mW/m}^2$$

Example



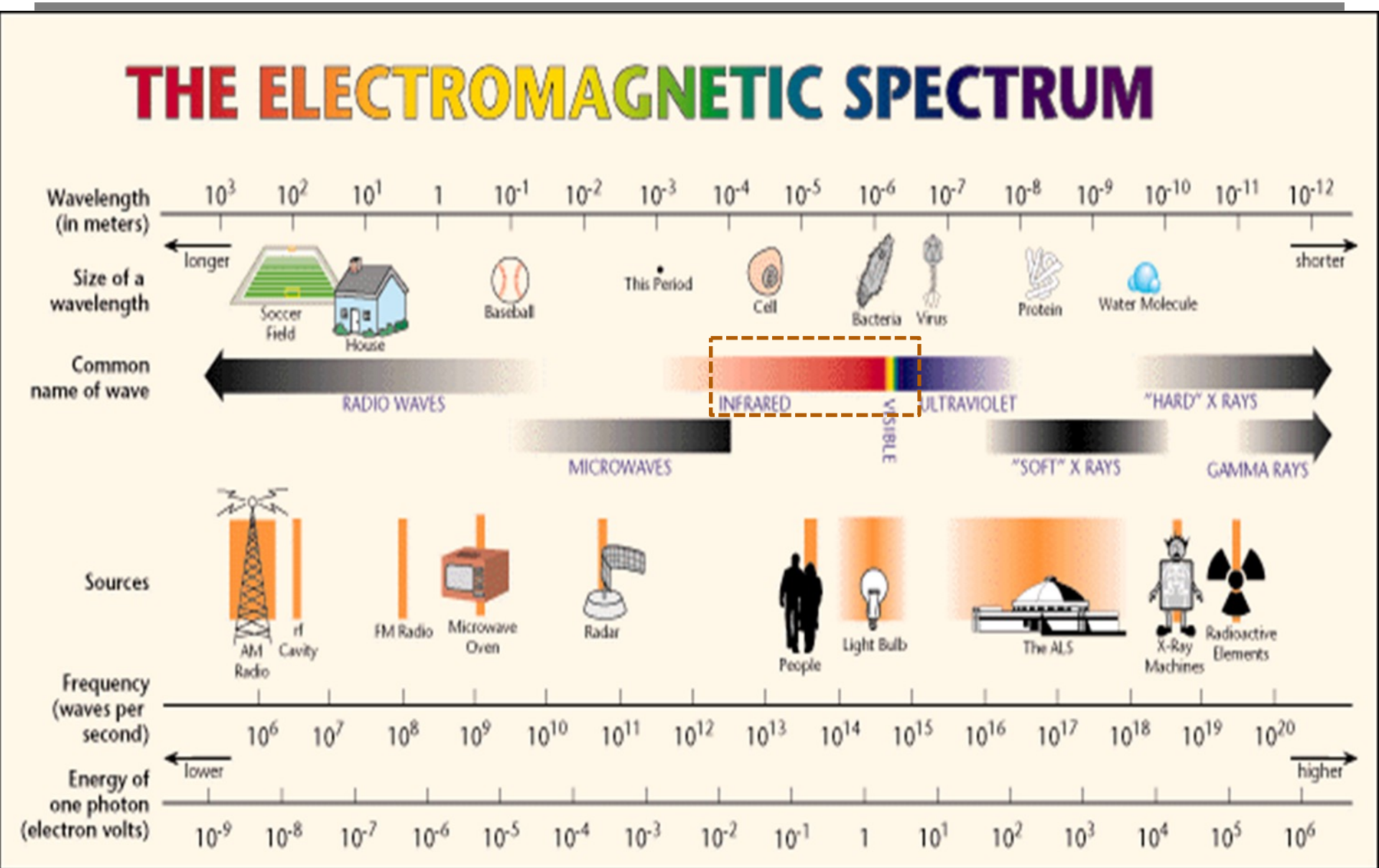
$$Total\ solar\ power = 3\times10^{26}\ W$$

$$Solar\ power\ density\ at\ Earch\ surface = \frac{3\times10^{26}}{4\pi r^2}$$

$$S_{ave} = \frac{{E_o}^2}{2|\eta|} \approx 1\ kW/m^2$$

$$E_o = 890\ V/m$$

Optical Spectrum



Q & A

Send me an email if you have any questions.